VISHAL GASTU

AI Engineer · Software Engineer — B.Tech CSE, SRM Institute of Science and Technology

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PROFESSIONAL SUMMARY

Computer Science undergraduate with hands-on experience in AI/ML, LLM engineering, and production-oriented software development. Experienced in fine-tuning and evaluating LLM systems with PyTorch, HuggingFace, TRL, GRPO, QLoRA, and on-device models, alongside building REST APIs and backend services in Go and Python. Currently researching runtime safety and compliance monitoring for AI agents (UROP, SRMIST) toward a publication, and focused on taking AI systems from experimentation to deployable, evaluable applications.

SKILLS

AI / LLM Engineering: Python, PyTorch, HuggingFace, TRL, GRPO, QLoRA, LLM Fine-Tuning, On-device LLMs, LLM Agents & Runtime Monitoring

Backend / Systems: Go, Python, REST APIs, PostgreSQL, SQL Optimization, JWT Auth, Layered Architecture

Machine Learning: Deep Learning, Reinforcement Learning, Transformers, Ensemble Learning, Model Evaluation

Software Engineering: Data Structures & Algorithms, OOP, Git, GitHub, Debugging, Testing

Frontend: React, Next.js, TypeScript, HTML, CSS

EXPERIENCE

Full Stack Engineer Intern — Qube Cinema Technologies Jul 2026 – Aug 2026 | Chennai, TN · Remote

- Built JWT-authenticated REST APIs in Go with PostgreSQL-backed reporting workflows for enterprise-grade cinema distribution software.
- Implemented dynamic filtering, search, sorting, pagination, and summary analytics over large datasets.
- Applied SQL optimization, indexing, complex joins, and parameterized queries to keep reporting fast and reliable at scale.
- Delivered Release and Mapped CPL reporting modules within a layered Handler–Service–Repository architecture, pairing the Go backend with a React/Next.js frontend.

AI / RESEARCH EXPERIENCE

UROP — Verifiable Observability: Runtime Safety Monitoring for AI Agents SRMIST · Ongoing

Python · LLM Agents · Runtime Monitoring · AI Safety

- Designing a runtime monitoring and enforcement framework that grades AI-agent reasoning and actions against SOPs and blocks unsafe actions before execution, across finance, healthcare, and code-execution domains.
- Architected a four-stage pipeline — Strategy Profiler (risk classification) → Rule Bank (SOPs) → Constraint Compliance Monitor (Allow/Flag/Block) → Metrics Engine — backed by a trajectory store for auditability.
- Built a literature survey spanning 20+ papers on LLM-agent safety, runtime enforcement, and constraint drift to position the work for publication.

PROJECTS

CRAFT — Adaptive Fine-Tuning for Small Language Models

Python · PyTorch · HuggingFace · TRL · GRPO · QLoRA · GGUF

- Built a four-phase fine-tuning framework combining GRPO-based reinforcement learning, QLoRA, step-level self-supervised contrastive learning, and adaptive curriculum learning.
- Replaced neural reward models with deterministic execution verifiers (a math interpreter and a logic classifier) to eliminate reward noise.
- On Phi-3-Mini (3.8B), delivered consistent accuracy gains over baseline — GSM8K 48%→62%, StrategyQA 42%→71%, MMLU 37%→63% — re-validated across multiple runs (±1%).
- Quantized the final model to a 4-bit GGUF (~2.2 GB) artifact for CPU-only inference.

Viren — Privacy-First Investment Intelligence

Flutter · Dart · Python · SQLite · On-device LLM

- Built an offline-capable app that reconstructs investment portfolios from broker emails and encrypted financial documents.
- Developed an end-to-end pipeline for email authentication, PDF decryption, trade extraction, and portfolio reconciliation.
- Integrated an on-device LLM for semantic repair of noisy financial document text, keeping sensitive data local.

RESEARCH PUBLICATION

Automated Liver Fibrosis Staging Using a Transformer-based Weighted Soft Voting Ensemble

2026 International Conference on Natural Language Processing and Computer Vision (ICNPCV) · Global Academy of Technology, Bengaluru · Presented July 17–18, 2026

EDUCATION

SRM Institute of Science and Technology, Kattankulathur, Tamil Nadu — B.Tech in Computer Science and Engineering — CGPA: 8.53/10.0 · 2024–2028

Class XII — Chettinad Vidyashram, CBSE: 80.2% (2024) · Class X — Velammal New Gen School, CBSE: 89% (2022)

ACHIEVEMENTS & CERTIFICATIONS

1st Place — The Last Commit hackathon (₹5,000 prize) · Finalist — Hack The Knight (36-hr) · Level 2 — Smart India Hackathon 2025 · NPTEL: Java Programming (93.7%), OOP (75%), DBMS (65%)